

ADITYA ANANTHARAMAN

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EDUCATION

Carnegie Mellon University (CMU), School of Computer Science Master of Computational Data Science (MCDS) <i>GPA: 4.11/4.33</i>	Pittsburgh, USA 2019 - 2020
National Institute of Technology Karnataka, Surathkal (NITK) Bachelor of Technology Information Technology <i>GPA: 9.54/10, Class Rank: 5/103</i>	Surathkal, India 2015 - 2019

EXPERIENCE

Applied Scientist II, Amazon	Feb 2021 - Present London, GB
- Led development and production deployment of a multimodal LLM-based brand-safety system in the ad-serving pipeline, reducing unsafe ad impressions alongside harmful content by 99%. - Led large-scale pre-training of MoE-based behavioral foundation models with up to 25B parameters on billions of user interaction sequences to boost click-through rate (CTR) prediction in e-commerce advertising. - Compressed large behavioral models by 350x using cross-architecture knowledge distillation, enabling deployment in low-latency ad-serving systems. - Proposed a Reverse Distillation (RD) training framework that enables training large encoder-based transformer models (up to 50 billion parameters) using noisy feedback signals (like clicks) by leveraging knowledge from smaller models. - Applied parameter-efficient LoRA fine-tuning to adapt LLMs for product retrieval and ranking applications in marketplace search and sponsored ads systems.	
Applied Scientist Intern, Amazon	May 2020 - Aug 2020 Seattle, USA
Developed co-teaching-based and BERT-based models to link context-of-use entities with products to improve search experience. Proposed approaches improved both precision and coverage of links compared to lexical matching.	
Research Intern, IIT-Hyderabad	Aug 2018 - Dec 2018 Hyderabad, India
Developed a novel multi-space model for Zero-Shot Object Detection (ZSD), achieving a 14% increase in mean Average Precision (mAP) compared to the state-of-the-art on Pascal VOC.	
Software Engineering Intern, Microsoft	May 2018 - July 2018 Hyderabad, India
Built a cluster management solution for the Azure Networking team, providing streamlined management of clusters.	

SELECTED PUBLICATIONS

- X Dong, **Aditya Anantharaman**, H Pugaliya, K Zhong “Multi-Scale Model Compression via Nested Matrix Learning”, LREC 2026 (Oral) [\[Paper\]](#)
- J Zhang, A Muhamed, **Aditya Anantharaman**, et al. “ReAugKD: Retrieval-Augmented Knowledge Distillation For Pre-trained Language Models”, ACL 2023 (Oral) [\[Paper\]](#) [\[Blog\]](#)
- M Yang*, **Aditya Anantharaman***, D C Robert*, Z Kitowski* “Graph Relation Transformer: Incorporating pairwise object features into the Transformer architecture”, Visual QA Workshop, CVPR 2021 [\[Paper\]](#) *Equal contribution
- D Gupta, **Aditya Anantharaman**, N Mangain, S Kamath, V Balasubramanian, C V Jawahar “A Multi-Space Approach to Zero-Shot Object Detection”, WACV 2020 [\[Paper\]](#)
- M Vikram, **Aditya Anantharaman**, Suhas BS, S Kamath, “An Approach for Multimodal Medical Image Retrieval using Latent Dirichlet Allocation”, India KDD CoDS-COMAD 2019 (Oral). Short paper at AI for Social Good Workshop, NeurIPS 2018 [\[Paper\]](#) [\[Poster\]](#)

SKILLS

Programming Languages:	Python, C++, C, Java, MySQL
Deep Learning:	PyTorch, TensorFlow, Deepspeed
Cloud Platforms and Tools:	AWS, Azure, Google Cloud Platform, Hadoop MapReduce, PySpark

ACADEMIC SERVICE

Reviewer: ACL 2024, EMNLP 2024, ES-FoMo-II Workshop ICML 2024